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Trincomalee Campus, Eastern University, Sri Lanka

Faculty of Applied Science

Department of Computer Science

Bachelor of Computer Science (2018/2019)

Year I Semester II Examination (March 2022)

CO 1225: Computer Architecture

Answer all questions **Time: 02 Hours**

Q1:

a) Convert the following hexadecimal numbers into decimal, binary and octal numbers.

[2*12=24 Marks]

i) AC5B4,

ii) 6C8DA.

b) Write the following decimal numbers in eight-bit two's complement, do the addition/subtraction, convert your answer back to decimal. **[3*16=48 Marks]**

i) -95+15,

ii) -82 +46,

iii) -122- 22.

c) Represent the following numbers in Binary Coded Decimal code. **[2*3=06 Marks]**

i) 789,

ii) 12483.

d) What are the disadvantages of using ASCII (American Standard Code for Information Interchange) over Unicode? **[12 Marks]**

e) Convert the following binary numbers into its decimal number equivalent. **[2*5=10 Marks]**

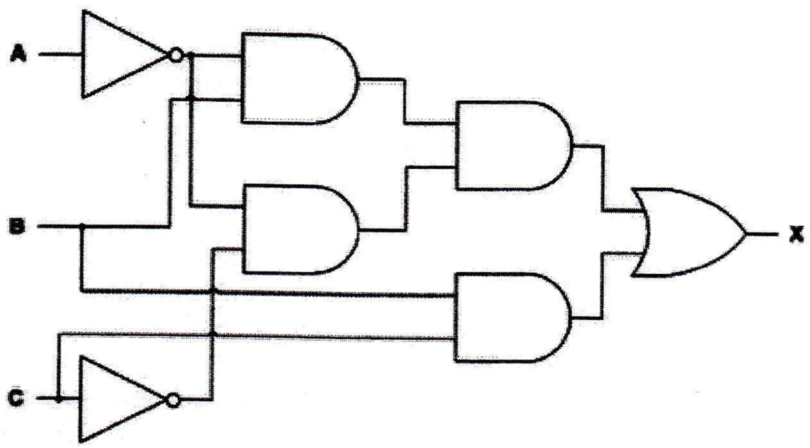
i) 0.1011_2,

ii) 1111.01001_2.

Q2:

a) Explain what is a combinational circuit and its characteristics. **[10 Marks]**

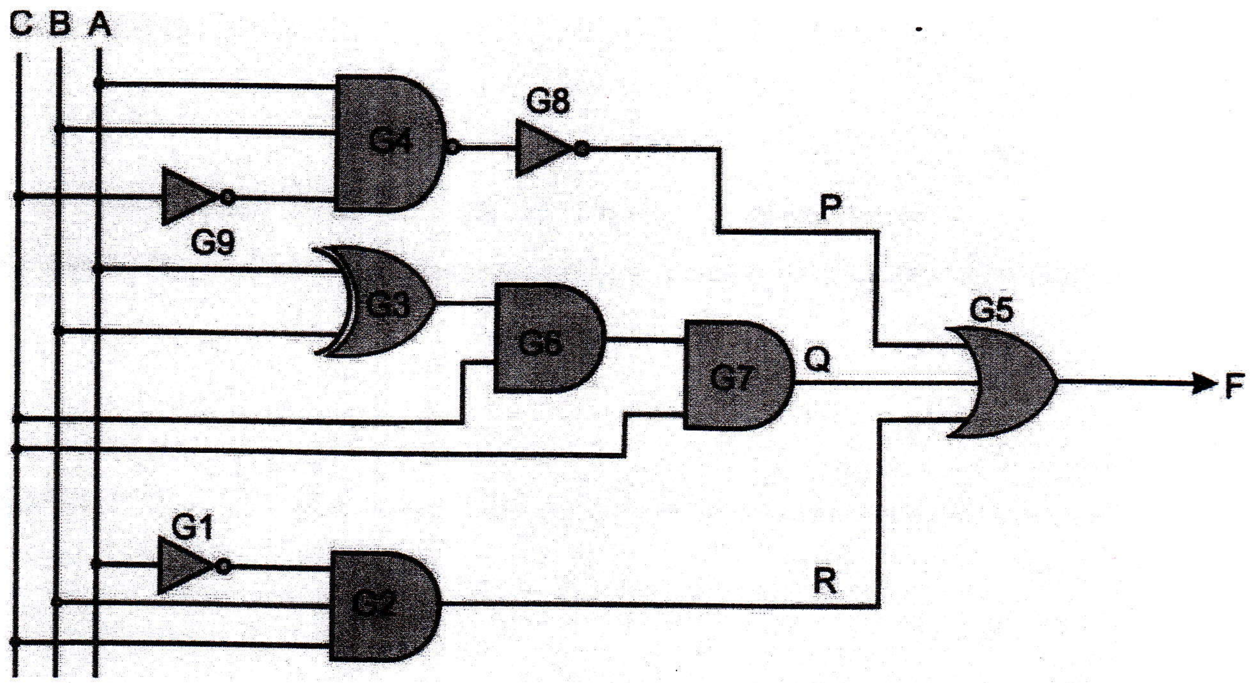
b) Consider the diagram below and represent the Boolean algebra for X. **[20 Marks]**



c) Draw a logic circuit for the following logic statement: **[25 Marks]**

X=1 if (A is 1 AND B is 1) OR (A is NOT 1 AND C is 1))

d) Consider the diagram below:



i) Construct a truth table for variable F with inputs A, B, C. **[25 Marks]**

ii) Write the Boolean algebra expression for output F and simplify it. **[20 Marks]**

Q3:

a) Write the machine language code for the following pseudo-code using the given memory locations. **[24 Marks]**

Pseudo-code:

Set X to Y * Z

Set P to Q / R

Then

Set K to X + P

Variable	Memory Location
X	789
Y	658
Z	690
P	1111
Q	1200
R	101
K	65

b) Explain the "little-endian system" concept using a suitable example. **[16 Marks]**

c) Compare and contrast the "Complex Instruction Set Computers (CISC)" versus "Reduced Instruction Set Computer (RISC)". **[24 Marks]**

d) Describe the advantages of single-core processors over multicore processors. **[12 Marks]**

e) Briefly explain the instruction execution cycle with a suitable diagram. **[24 Marks]**

Q4:

a) Describe the difference between the modern solid-state drive (SSD) and the hard disk drive (HDD). List down the advantages and disadvantages of the SSD over the HDD. **[24 Marks]**

b) Describe the following addressing modes with suitable diagrams. **[24 Marks]**

i) Register indirect addressing

ii) Immediate addressing

c) Explain the Von Neumann architecture with a suitable diagram. **[24 Marks]**

d) The table shows a segment of primary memory from a Von Neumann model computer. **[12 Marks]**

Address	Contents
10001	11001101
10010	11110001
10011	10101111
10100	10000110
10101	00011001
10110	10101100

The program counter contains the data 10010.

i) State the data that will be placed in the memory address register (MAR).

ii) State the data that will be placed in the memory data register (MDR).

e) Describe the stored program concept when applied to the Von-Neumann model.

[16 Marks]